FAILURE OF GEOMETRIC REDUCEDNESS OF THE GENERIC FIBRE PERSISTS OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7180-7186 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7188-7212, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Geometric reducedness of fibres and the spreading of geometric reducedness from the generic fibre; the argument is retained.

Source reference. Original lemma lemma-base-change-fibres-geometrically-reduced: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7161–7169. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-make-generic-fibre-geometrically-reduced: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6719–6739. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-nonreduced-in-neighbourhood: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7110–7115. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0384: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7212–7212. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume Y irreducible with generic point η and f of finite type. If X_η is not geometrically reduced, then there exists a nonempty open $V \subset Y$ such that for all $y \in V$ the fibre X_y is not geometrically reduced.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Apply Lemma lemma-make-generic-fibre-geometrically-reduced to get

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & X_V & \longrightarrow & X \\ \downarrow{\scriptstyle f'} & & \downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & V & \longrightarrow & Y \end{array}$$

with all the properties mentioned in that lemma. Let η' be the generic point of Y'. Consider the morphism $X' \to X_{Y'}$ (which is the reduction morphism) and the resulting morphism of generic fibres $X'_{\eta'} \to X_{\eta'}$. Since $X'_{\eta'}$ is geometrically reduced, and X_η is not this cannot be an isomorphism, see Varieties, Lemma 0384. Hence $X_{\eta'}$ is nonreduced. Hence by Lemma lemma-nonreduced-in-neighbourhood the fibres of $X_{Y'} \to Y'$ are nonreduced at all points $y' \in V'$ of a nonempty open $V' \subset Y'$. Since $g : Y' \to V$ is a homeomorphism Lemma lemma-base-change-fibres-geometrically-reduced proves that $g(V')$ is the open we are looking for. $\square$